

Late Delivery Risk Prediction in Logistics Using Machine Learning

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Abstract—Late delivery is a major challenge in logistics and supply chain management. This paper presents a machine learning based late delivery risk prediction system using Random Forest classification. The system analyzes shipment, customer, and order related attributes to identify deliveries that are likely to be delayed. An interactive dashboard was also developed for visualization and real-time risk monitoring.

Index Terms—Machine Learning, Logistics, Random Forest, Supply Chain Analytics, Late Delivery Prediction

I. INTRODUCTION

Late deliveries create operational challenges in logistics organizations and negatively impact customer satisfaction. Traditional systems usually detect delays after they occur, making it difficult for managers to take preventive action.

Machine learning enables organizations to identify risky shipments before delivery completion. By analyzing historical shipment data, predictive models can estimate delay probability and help logistics teams prioritize critical shipments.

This project develops a machine learning based late delivery prediction system using Random Forest and an interactive Streamlit dashboard.

II. LITERATURE REVIEW

Several researchers have explored the use of machine learning techniques in supply chain and logistics management. Random Forest, Decision Trees, Support Vector Machines, and Gradient Boosting algorithms have been widely used for predictive analytics in transportation systems.

Researchers have also applied predictive analytics for shipment tracking, warehouse optimization, route planning, and customer demand forecasting. These studies demonstrate that machine learning can significantly improve operational efficiency and reduce transportation costs.

Although previous studies focused on logistics optimization, there remains a need for practical systems capable of predicting late delivery risk before shipment completion. The proposed work addresses this gap by combining machine learning prediction with an interactive business dashboard.

III. PROBLEM STATEMENT

APL Logistics faces challenges such as unpredictable shipment delays, increased operational costs, and limited ability to identify high-risk orders. The company requires a predictive solution that can identify potential delivery delays before shipment completion.

IV. OBJECTIVES

- Analyze logistics shipment data.
- Build a machine learning model for delay prediction.
- Evaluate model performance using classification metrics.
- Develop an interactive dashboard for business users.
- Provide real-time shipment risk prediction.

V. DATASET DESCRIPTION

The dataset contains shipment, customer, sales, and delivery related information. Important attributes include payment type, shipping duration, scheduled shipment days, sales value, order benefit, shipping mode, and delivery status.

VI. SYSTEM ARCHITECTURE

The proposed system consists of five major components: data collection, preprocessing, model training, prediction engine, and dashboard visualization.

The data collection layer gathers logistics and shipment information from historical records. The preprocessing layer cleans and transforms the raw dataset into a machine learning compatible format. The model training layer uses the processed dataset to build a Random Forest classifier capable of predicting delivery risk.

The prediction engine receives shipment-related inputs and generates risk scores. Finally, the dashboard visualization layer displays key performance indicators, delay trends, and prediction results in an interactive format for business users.

This architecture ensures scalability, reliability, and ease of deployment in real-world logistics environments.

TABLE I
IMPORTANT DATASET ATTRIBUTES

Attribute	Description
Type	Payment Method
Days for Shipping (Real)	Actual Shipping Days
Days for Shipment (Scheduled)	Planned Shipping Days
Benefit per Order	Profit from Order
Sales per Customer	Customer Sales Value
Shipping Mode	Delivery Method
Delivery Status	Final Shipment Status

VII. METHODOLOGY

The proposed system follows a machine learning workflow consisting of data preprocessing, exploratory data analysis, model training, evaluation, and deployment.

A. Data Preprocessing

The dataset was cleaned by handling missing values, removing irrelevant features, and encoding categorical variables. Numerical and categorical features were prepared for machine learning training.

B. Feature Engineering

Important shipment and order-related variables were selected to improve model performance. Feature engineering helped improve prediction quality and reduce noise.

KPI Summary			
Total Orders	Delayed Orders	Delay Rate	Avg Shipping Days
30,000	16,462	54.87%	3.5
Total Sales	Total Profit	Sales at Risk	Profit at Risk
\$6,130,348	\$664,321	\$3,356,414	\$355,656

Fig. 1. Dashboard KPI section showing overall logistics performance

C. Data Cleaning and Transformation

Data cleaning is a critical step in machine learning projects because logistics datasets often contain missing values, duplicate records, inconsistent formats, and irrelevant attributes. In this project, missing values were handled appropriately, unnecessary columns were removed, and categorical variables were encoded into numerical representations suitable for machine learning algorithms.

The processed dataset was further standardized to ensure consistency across features. Data transformation techniques helped improve model learning and reduced the impact of noisy records. The resulting dataset was then used for exploratory data analysis and model training.

D. Model Training Process

The Random Forest classifier was trained using historical shipment records. The dataset was divided into training and testing subsets to evaluate model generalization performance. Multiple decision trees were created during training, and the final prediction was generated through majority voting.

Random Forest was selected because it reduces overfitting, handles high-dimensional data effectively, and provides feature importance information. These characteristics make it highly suitable for logistics risk prediction tasks.

VIII. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted to understand delivery patterns and identify factors contributing to shipment delays.

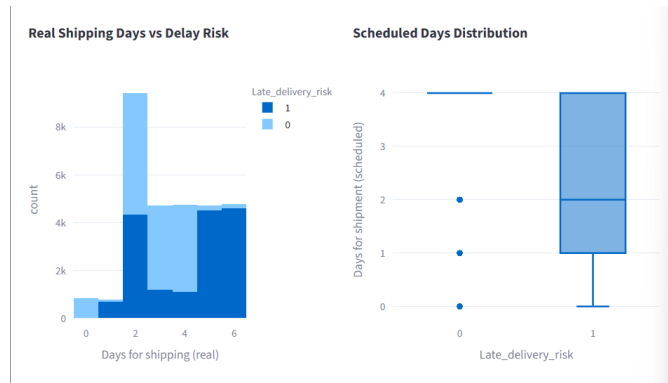


Fig. 2. Distribution of delivery status

The analysis showed the proportion of delayed and on-time shipments and helped identify class distribution within the dataset.

Sales by Shipping Mode



Fig. 3. Shipping mode wise late delivery analysis

Shipping mode analysis provided insights into which shipment methods contributed most to delivery delays.

IX. MODEL DEVELOPMENT

A Random Forest classifier was selected for late delivery prediction because of its robustness, ability to handle mixed data types, and resistance to overfitting.

The dataset was divided into training and testing subsets. The model learned shipment patterns from historical records and classified deliveries as either delayed or on-time.

Random Forest combines multiple decision trees to improve prediction accuracy and generate reliable classifications.

X. MODEL EVALUATION

The model was evaluated using Accuracy, Precision, Recall, and F1-Score. These metrics provide a comprehensive understanding of classification performance.

TABLE II
MODEL EVALUATION METRICS

Metric	Value
Accuracy	Add Your Value
Precision	Add Your Value
Recall	Add Your Value
F1-Score	Add Your Value

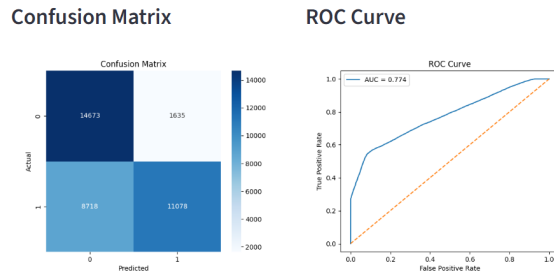


Fig. 4. Confusion matrix of the Random Forest classifier

The confusion matrix illustrates the model's ability to distinguish between delayed and on-time deliveries.

XI. PREDICTION DASHBOARD

An interactive Streamlit dashboard was developed to provide logistics insights and real-time risk prediction. Users can enter shipment details and receive an instant prediction indicating whether a shipment is likely to be delayed.

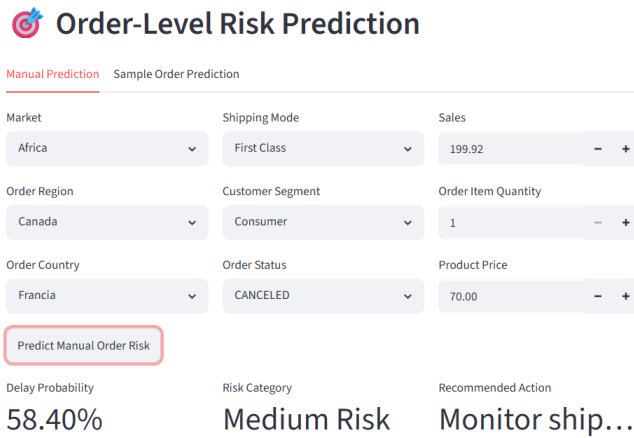


Fig. 5. Real-time late delivery risk prediction output

The dashboard improves usability by transforming machine learning outputs into actionable business insights.

XII. RESULTS AND DISCUSSION

The developed Random Forest model demonstrated the capability to identify late delivery risks using historical shipment information. The model successfully learned patterns associated with delayed deliveries and generated meaningful predictions for unseen shipment records.

The confusion matrix indicated that the classifier was able to distinguish between delayed and non-delayed shipments with satisfactory performance. The evaluation metrics such as accuracy, precision, recall, and F1-score provided quantitative evidence of the model's effectiveness.

The dashboard further enhanced the usefulness of the system by presenting operational insights through visual analytics. Managers can use these insights to identify risky shipments, monitor delivery performance, and make proactive decisions.

The prediction module allows users to enter shipment details and obtain an estimated delay risk score. This functionality enables logistics teams to take preventive actions before shipment delays occur, reducing operational costs and improving customer satisfaction.

XIII. BUSINESS IMPACT

The proposed system provides significant value to logistics organizations by enabling proactive risk management. Instead of reacting to delivery failures after they occur, managers can identify high-risk shipments in advance and take corrective measures.

The system supports better shipment prioritization, warehouse planning, route management, and customer communication. By reducing delivery delays, organizations can improve customer satisfaction, strengthen service reliability, and lower operational expenses.

XIV. CONCLUSION

This project presented a machine learning based late delivery prediction system using Random Forest classification. The developed solution provides risk prediction, business insights, and real-time monitoring through an interactive dashboard.

The results show that predictive analytics can significantly improve logistics planning and operational efficiency.

XV. FUTURE SCOPE

Several enhancements can be incorporated into the proposed system in future work. Real-time shipment tracking data can be integrated to improve prediction accuracy and provide dynamic risk assessment. Additional external factors such as weather conditions, traffic congestion, fuel costs, and regional transportation constraints can also be included.

Advanced machine learning algorithms such as XGBoost, LightGBM, Gradient Boosting, and Deep Learning models may further improve predictive performance. Explainable AI techniques can be integrated to provide transparent explanations behind delay predictions.

The system can also be deployed as a cloud-based enterprise application and integrated directly with logistics management platforms to enable automated alerts and intelligent shipment recommendations.

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